

Name: _____

Exam Style Questions

Laws of Indices



Equipment needed: Pen, Calculator

Guidance

1. Read each question carefully before you begin answering it.
2. Check your answers seem right.
3. Always show your workings

Video Tutorial

www.corbettmaths.com/contents

Videos 17, 174



Answers and Video Solutions



1.



$$2^2 \times 2^3 = 2^{\boxed{}}$$

Write the missing power in the box.

(1)

2.

Simplify $5^4 \times 5^3$



Circle the correct answer.

25^7

5^{12}

5^7

25^{12}

(1)

3.

Simplify $n^2 \times n^4$



.....
(1)

4.

$$p^6 \times p^2 = p^x$$



Find the value of x

.....
(1)

5.

Jonah was asked to simplify $y^8 \times y^3$



His answer is y^{24}

Explain Jonah's mistake.

.....

.....

(1)

6.



(a) Simplify

$$w^4 \times w^4$$

.....
(1)

(b) Simplify

$$a^6 \div a^3$$

.....
(1)

7.



(a) Simplify

$$m^5 \times m^3$$

.....
(1)

(b) Simplify

$$m^8 \div m^2$$

.....
(1)

(c) Simplify

$$(m^3)^2$$

.....
(1)

8. Simplify $\frac{c^{12}}{c^4}$



.....
(1)

9. Simplify $\frac{m^9 \times m}{m^5}$



.....
(2)

10. Match the expressions.



$$c^{10} \div c^5$$

$$c^{16}$$

$$c^7 \times c^3$$

$$c^5$$

$$(c^8)^2$$

$$c^2$$

$$\frac{c^6}{c^4}$$

$$c^{10}$$

(4)

11. Simplify the following.



$$\frac{s^3 \times s^4}{s^2}$$

.....
(2)

12.



(a) Simplify $w^3 \times w^{-5}$

.....
(1)

(b) Simplify $a^4 \div a^{-2}$

.....
(1)

13. Simplify $c^7 \times (c^6)^4$



.....
(2)

14. Work out the value of $\frac{2^9 \times 2^{-2}}{2^3}$



.....
(2)

15. Simplify $(5^{-3} \times 5^7)^3$
-  Give your answer as a power of 5.

.....
(2)

16. Simplify $2a^3c^3 \times 3a^2c$



.....
(2)

17. Simplify



$$\frac{10m^5n^4}{2m^2n}$$

.....
(2)

18. Simplify $(7x^3)^2$



Circle the correct answer.

$49x^5$

$14x^6$

$14x^5$

$49x^6$

.....
(1)

19. Simplify $(2m^4)^3$



.....
(2)

20. Simplify $(2xy^3)^4$



.....
(2)

21. Work out the value of $3^{16} \div (3^2)^7$



.....
(2)

22. Simplify



$$\frac{a^{\frac{1}{5}} \times a^{\frac{2}{3}}}{a^{\frac{3}{5}}}$$

.....
(3)

23. Simplify fully



$$\sqrt{\frac{50\pi a^5}{2\pi a^3}}$$

.....
(3)

24. Work out



$$\sqrt[3]{\frac{2^3 \times 3^8}{3^2}}$$

.....
(2)

25. Simplify $25 \times 5^3 \times 5^6$



Give your answer as a power of 5.

.....
(2)

26. Write 27×9^x as a power of 3 in terms of x



.....
(3)